

Generic Upgrade Tool – Support Knowledge Base

Fictional training knowledge base for a GraphRAG client-support assistant. All product names, clients, environments, versions, incidents, and procedures are fictional.

1. Product Overview

Upgrade Tool is a fictional deployment and application-upgrade utility used by support engineers to prepare, validate, execute, and verify application upgrades across client environments.

Core components

- Upgrade Manager – controls upgrade jobs and workflow state.
- Precheck Engine – validates prerequisites before an upgrade.
- Package Repository – stores approved application packages.
- Configuration Service – supplies environment-specific configuration.
- Execution Agent – runs upgrade actions on the target environment.
- Health Monitor – performs post-upgrade validation.

2. Environment Model

Each client is associated with one or more environments. An environment contains an application version, database connection, configuration profile, and execution agent.

Environment	Purpose	Typical Upgrade Window
Development	Engineering validation	Flexible
Staging	Pre-production validation	Scheduled
Production	Live client workload	Approved maintenance window

3. Fictional Client Relationships

- Northstar Health Group uses the Production environment and Upgrade Tool version 4.2.
- Greenfield Care Network uses Staging and Production environments.
- Riverside Medical Services uses the Development environment for upgrade validation.
- Northstar Health Group and Greenfield Care Network use the same approved package repository.
- Greenfield Care Network uses Configuration Profile B.

4. Standard Upgrade Workflow

- 1. Confirm request:** Verify client, environment, target version, maintenance window, and approval.
- 2. Backup:** Confirm the latest application/database backup is available according to the environment procedure.
- 3. Run prechecks:** Validate disk space, service status, connectivity, supported version, configuration, and package availability.
- 4. Obtain package:** Download the approved package from the Package Repository and verify its checksum.
- 5. Execute upgrade:** Run the upgrade through Upgrade Manager and monitor the Execution Agent.

- 6. Validate services:** Confirm application services, database connectivity, login, and critical workflows.
- 7. Review logs:** Check upgrade and application logs for errors or warnings.
- 8. Close request:** Record the final version, validation results, timestamps, and follow-up actions.

5. Precheck Troubleshooting

Insufficient disk space

Symptom: Upgrade package cannot be unpacked or temporary files cannot be created.

Recommended action: Check available disk space, remove approved temporary files, then rerun prechecks.

Package unavailable

Symptom: Upgrade Manager cannot find the requested target package.

Recommended action: Confirm target version and repository availability. Do not use an unapproved package.

Agent offline

Symptom: Execution Agent does not respond to Upgrade Manager.

Recommended action: Check agent service status, host connectivity, and agent logs.

Database connectivity failure

Symptom: Precheck cannot establish a database connection.

Recommended action: Verify configured endpoint, credentials, network connectivity, and database service status.

Unsupported source version

Symptom: Current application version is outside the supported upgrade path.

Recommended action: Confirm the supported upgrade path and perform any required intermediate upgrade.

Configuration mismatch

Symptom: Environment configuration does not meet package requirements.

Recommended action: Compare the environment profile with target version requirements and correct approved settings.

6. Common Upgrade Failures

Upgrade stops during package extraction

Possible cause: Package may be incomplete or checksum validation may fail.

Action: Verify checksum, download the approved package again, and review extraction logs.

Upgrade completes but application is unavailable

Possible cause: A service may not have restarted correctly.

Action: Check service status, application logs, port availability, and dependent services.

Users receive login errors after upgrade

Possible cause: Authentication configuration or service dependencies may be incorrect.

Action: Validate authentication configuration and review application and authentication logs.

Database migration fails

Possible cause: Database permissions, connectivity, or migration state may be incorrect.

Action: Stop further upgrade actions, preserve logs, verify database access, and follow the database recovery procedure.

Health check fails

Possible cause: One or more application dependencies may not be healthy.

Action: Review health-monitor details and validate each dependent service.

7. Support Escalation

Escalate when the issue cannot be resolved using the documented procedure, when the upgrade path is unsupported, or when a production upgrade has an unresolved failure.

- Collect client and environment name.
- Record current and target application versions.
- Capture upgrade job ID and timestamps.
- Attach relevant Upgrade Manager, Execution Agent, application, and database logs.
- Record the exact error message and the last successful step.
- Document troubleshooting steps already attempted.

8. Graph Relationships for Support Knowledge

- Client → USES_ENVIRONMENT → Environment
- Environment → RUNS_VERSION → Application Version
- Environment → HAS_CONFIGURATION → Configuration Profile
- Environment → USES_DATABASE → Database
- Upgrade Package → TARGETS_VERSION → Application Version
- Upgrade Job → EXECUTES_ON → Environment
- Upgrade Job → USES_PACKAGE → Upgrade Package
- Precheck → VALIDATES → Environment
- Client → HAS_UPGRADE_JOB → Upgrade Job

9. Example Questions for GraphRAG Testing

- What should a support engineer check if an upgrade package cannot be found?
- Which environment does Northstar Health Group use?
- What checks should be completed before executing an upgrade?
- What information should be collected before escalating a failed production upgrade?
- What can cause a database migration to fail?
- Which clients use the same package repository?
- What relationships exist between an upgrade job, package, and environment?

- What should be checked if the application is unavailable after a successful upgrade?
- Which configuration profile is used by Greenfield Care Network?
- What is the standard upgrade workflow?

10. Demo Safety Note

This document is intentionally fictional and contains no real company, client, credential, infrastructure, or proprietary operational information. It is suitable as a demonstration dataset for the GraphRAG project.